



PRODUCT NAME
Metrolex Injection

NAME AND STRENGTH OF ACTIVE SUBSTANCE
Each 100 mL contains 500 mg metronidazole

PRODUCT DESCRIPTION
Clear, colourless sterile solution.

PHARMACODYNAMICS
Pharmacotherapeutic group: antibacterials for systemic use: imidazole derivatives & antiprotozoals: nitroimidazole derivatives
Mechanism of Action
Metronidazole is an anti-infectious drug belonging to the pharmacotherapeutic group of nitroimidazole derivatives, which have effect mainly on strict anaerobes. This effect is probably caused by interaction with DNS and different metabolites.
Metronidazole has antibacterial and antiprotozoal actions and is effective against anaerobic bacteria and against *Trichomonas vaginalis* and other protozoa including *Entamoeba histolytica* and *Giardia lamblia*.

Microbiological Susceptibility
The prevalence of acquired resistance may vary geographically and with time for selected species and local information is desirable, particularly when treating severe infections. Metronidazole is usually active against the following microorganisms (just an approximate guidance):

SUSCEPTIBLE		
<u>Gram-Negative Aerobes:</u> <i>Helicobacter pylori</i>		
<u>Anaerobes:</u>		
<i>Bacteroides fragilis</i>	<i>Clostridium difficile</i>	<i>Peptostreptococcus</i>
<i>Bifidobacterium</i> ->-resistant (70%)	<i>Clostridium perfringens</i>	<i>Prevotella</i>
<i>Blifophila</i>	<i>Eubacterium</i>	<i>Porphyromonas</i>
<i>Clostridium</i>	<i>Fusobacterium</i>	<i>Veillonella</i>
RESISTANT		
<u>Gram-Positive Aerobes:</u> <i>Actinomyces</i>		
<u>Anaerobes:</u> <i>Mobiluncus</i> <i>Propionibacterium acnes</i>		
ANTIPARASITIC ACTIVITY		
<i>Entamoeba histolytica</i> <i>Giardia intestinalis</i> <i>Trichomonas vaginalis</i>		

PHARMACOKINETICS
Distribution
After administration of a single 500 mg dose, mean peak plasma concentrations of metronidazole (C_{max} = 14 - 18 µg/mL) are reached at the end of a 20 minute infusion. Steady state plasma concentrations (C_{ss} = 17 and 13 µg/mL) are reached after administration of metronidazole every 8 or 12 hours respectively. Plasma protein binding is less than 10%.
Metabolism
Metronidazole is metabolized in the liver by hydroxylation, oxidation and glucuronidation. The major metabolites are acetic acid and 2-hydroxy-metabolite.
Elimination
More than 50% of the administered dose is excreted in the urine, as unchanged metronidazole and its metabolites. About 20% of the dose is excreted with faeces. The plasma elimination half-life of metronidazole was estimated to be 8 hours and 10 hours for 2-hydroxy-metabolite.
Hepatic Impairment
In patients with impaired liver function, the metabolism of metronidazole is expected to decrease, leading to an increase in the plasma elimination half-life. In patients with severe liver impairment, clearance may be decreased up to 65%, resulting in drugs accumulation in the body.
Renal Impairment
The plasma elimination half-life of metronidazole is not influenced by renal impairment. However, this may be increased for acetic acid and 2-hydroxy-metabolite. In the case of haemodialysis, metronidazole is rapidly excreted and the plasma elimination half-life is decreased 2.5 hours. Peritoneal dialysis does not appear to affect the elimination of metronidazole or its metabolites compared to patients with renal impairment.
INDICATION
■ For anaerobic bacteria infection especially *Bacteroides fragilis* and other species of bacteroides, Fusobacteria, Eubacteria, Clostridia and anaerobic cocci
■ For the treatment of septicaemia, bacteremia, brain abscess, necrotizing pneumonia, osteomyelitis, puerperal sepsis, pelvic abscess, pelvic cellulites, peritonitis and postoperative anaerobic infection
■ Prevention of postoperative infections due to anaerobic bacteria particularly species of bacteroides and anaerobic streptococcus

RECOMMENDED DOSAGE
General Dosage Recommendations
Adults and Children (>12 years)
■ Intravenous infusion of metronidazole 500 mg every 8 hours (infusion rate: 5 mL/min)
■ Metronidazole can be administered alone or in combination with other antibacterial agents (given separately)
■ It should be substituted by oral metronidazole 400 mg 3 times daily for 7 days as soon as feasible
Children (<12 years)
■ Intravenous infusion of metronidazole 7.5 mg/kg body weight (1.5 mL) or by oral 7.5 mg/kg body weight every 8 hours
Prophylaxis
Adults and Children (>12 years)
■ Intravenous infusion of metronidazole 500 mg before operation and continued every 8 hours until oral dose is obtainable. The oral dose of metronidazole is 200 - 400 mg 3 times daily for 7 days
Children (<12 years)
■ Intravenous infusion of metronidazole 7.5 mg/kg body weight (1.5 mL) or by oral 3.7 - 7.5 mg/kg body weight every 8 hours
Special Populations
Elderly
■ Dosage adjustment may be necessary
■ Plasma metronidazole levels should be monitored
Hepatic Impairment
■ Reduce doses below those usually recommended
■ Monitor plasma metronidazole levels and observe for toxicity
Renal Impairment
■ Dose adjustment is not necessary in patients with mild to moderate renal function impairment.
■ Do not specifically reduce the dose in anuric patients, since accumulated metabolites may be rapidly removed by dialysis
■ Unchanged drug and its metabolites are removed during haemodialysis and a further dose of metronidazole should be given after haemodialysis to replace the loss
ROUTE OF ADMINISTRATION
For intravenous infusion
CONTRAINDICATIONS
Hypersensitivity to metronidazole or to other imidazole derivatives
WARNINGS AND PRECAUTIONS
Liver Disease
Cases of severe hepatotoxicity/acute hepatic failure, including cases with a fatal outcome with very rapid onset after treatment initiation in patients with Cockayne syndrome have been reported with products containing metronidazole for systemic use. In this population, metronidazole should therefore be used after careful benefit-risk assessment and only if no alternative treatment is available. Liver function tests must be performed just prior to the start of therapy, throughout and after end of treatment until liver function is within normal ranges, or until the baseline values are reached. If the liver function tests become markedly elevated during treatment, the drug should be discontinued.
Patients with Cockayne syndrome should be advised to immediately report any symptoms of potential liver injury to their physician and stop taking metronidazole.
Active Central Nervous System (CNS) Disease
Metronidazole should be used with caution in patients with active disease of the peripheral and central nervous system. Severe neurological disturbances (including seizures, peripheral and optic neuropathies) have been reported. Stop metronidazole treatment if any abnormal neurologic symptoms occur such as ataxia, dizziness, confusion, etc. Encephalopathy has been reported in association with cerebellar toxicity characterized by ataxia, dizziness, dysarthria, and accompanied by CNS lesions seen on magnetic resonance imaging (MRI). The CNS lesions and symptoms are generally reversible within days to weeks upon discontinuation of metronidazole.
Intensive/Prolonged Metronidazole Therapy
The usual duration of therapy with intravenous metronidazole or other imidazole derivatives is usually less than 10 days. Limiting the duration of treatment is necessary because damage to human germ cells cannot be excluded. Intensive or prolonged therapy should be conducted only by specialist direction and under conditions of close surveillance, whereby the physician should bear in mind the possibility of peripheral neuropathy or leucopenia (usually reversible). In case of prolonged treatment, occurrence of undesirable effects such as paraesthesia, ataxia, dizziness and convulsive crises should be checked. High dose regimes have been associated with transient epileptiform seizures.
Blood Dyscrasias
Metronidazole should be used with caution in patients with evidence or history of blood dyscrasia as agranulocytosis, leukopenia and neutropenia have been observed following metronidazole administration.
Renal Disease
Metronidazole is removed during haemodialysis and should be administered after the procedure is finished. Patients with renal impairment, including patients receiving peritoneal dialysis, should be monitored for signs of toxicity due to the potential accumulation of toxic metronidazole metabolites.
Alcohol
Patients should be advised to discontinue consumption of alcoholic beverages or alcohol-containing products before, during, and up to 72 hours after taking metronidazole

Actual Size 100 %

because of a disulfiram-like effect (abdominal cramps, nausea, headaches, flushing, vomiting and tachycardia).

EFFECTS ON THE ABILITY TO DRIVE AND USE MACHINES

No studies have been performed following intravenous treatment with metronidazole on the ability to drive and use machines. Some adverse reactions to metronidazole such as seizure, dizziness, optic neuropathy, may impair the ability to drive or operate machines (See *Undesirable Effects*). Therefore it is recommended that patients should not drive or use machines.

INTERACTIONS WITH OTHER MEDICAMENTS

Disulfiram

Concurrent use of metronidazole and disulfiram may result in psychotic reactions and confusion. Metronidazole should not be given to patients who have taken disulfiram within the last two weeks.

Oral Anticoagulants

Metronidazole may increase the anticoagulant effects of warfarin and other oral anticoagulants, resulting in a prolongation of the prothrombin time and increased risk of haemorrhage. Patients taking metronidazole and warfarin or other oral coumarins concomitantly should have their prothrombin time and international normalized ratio (INR) monitored more frequently. Patients should be monitored for signs and symptoms of bleeding.

Lithium

Drug retention of lithium accompanied by evidence of possible renal damage has been reported in patients treated concurrently with metronidazole. Lithium treatment should be tapered or withdrawn before administering metronidazole. Plasma concentrations of lithium, creatinine and electrolytes should be monitored in patients taking lithium and metronidazole concomitantly.

Cimetidine

Cimetidine decreases the microsomal liver enzyme activity when used simultaneously with metronidazole. The reduction of drug metabolism and plasma clearance may result in metronidazole toxicity.

Busulfan

Plasma concentrations of busulfan may increase during concomitant treatment with metronidazole, which can result in serious busulfan toxicity such as sinusoidal obstruction syndrome, gastrointestinal mucositis, and hepatic veno-occlusive disease.

Phenytoin/Barbiturates

Concomitant administration of drugs that induce microsomal liver enzyme activity, such as phenytoin or phenobarbital, may accelerate the elimination of metronidazole and therefore decrease its efficacy.

Investigations

Metronidazole may interfere with serum aspartate transaminase (AST), alanine transaminase (ALT), lactate dehydrogenase (LDH), triglycerides, and glucose hexokinase determinations. Metronidazole causes an increase in ultraviolet absorbance at 340 nm resulting in falsely decreased values.

PREGNANCY AND LACTATION

Pregnancy

Metronidazole crosses the placental barrier. Clinical data on a large number of exposed pregnancies and animal data did not show teratogenic or foetotoxic effect. However unrestricted administration of nitroimidazole to the mother may be associated with carcinogenic or mutagenic risk for the unborn or newborn child. Therefore, metronidazole should not be given during pregnancy unless clearly necessary.

Lactation

Metronidazole is excreted in breast milk. During lactation either breast-feeding or metronidazole should be discontinued.

UNDESIRABLE EFFECTS

Summary of Safety Profile

The adverse reactions presented below are classified by system organ and frequency in decreasing order of the seriousness.

Frequency, type and severity of adverse reactions in children are the same as in adults.

System Organ	Frequency	Adverse Reactions
Blood and lymphatic system disorders	Uncommon	: Leukopenia
	Rare	: Agranulocytosis, pancytopenia, thrombocytopenia, neutropenia
	Not known	: Eosinophilia
Immune system disorders	Rare	: Anaphylactic shock, Jarisch-Herxheimer reaction
	Not known	: Hypersensitivity
Metabolism and nutrition disorders	Not known	: Decreased appetite
Psychiatric disorders	Rare	: Hallucinations
	Not known	: Depression, confusional state, insomnia
Nervous system disorders	Common	: Dysgeusia
	Uncommon	: Headache
	Rare	: Encephalopathy, meningitis aseptic, somnolence, seizure, neuropathy peripheral, dizziness, ataxia
	Not known	: Dysarthria, hypoaesthesia, paraesthesia
Eye disorders	Rare	: Optic neuropathy, diplopia, myopia
Cardiac disorders	Not known	: Tachycardia, palpitations

System Organ	Frequency	Adverse Reactions
Respiratory, thoracic and mediastinal disorders	Not known	: Dyspnoea
Gastrointestinal disorders	Common	: Glossitis, stomatitis, dry mouth
	Rare	: Pancreatitis, abdominal pain upper, diarrhoea, nausea, vomiting
	Not known	: Constipation, tongue discoloration
Hepatobiliary disorders	Rare	: Jaundice cholestatic
Skin and subcutaneous disorders	Rare	: Stevens-Johnson syndrome, erythema multiforme, angioedema, toxic epidermal necrolysis
	Not known	: Pruritus, swelling face, urticarial, hyperhidrosis, rash
Musculoskeletal and connective tissue disorders	Common	: Myalgia
	Not known	: Muscle spasms, arthralgia
Renal and urinary disorders	Rare	: Chromaturia
	Not known	: Dysuria
General and administration site conditions	Common	: Asthenia
	Rare	: Mucosal inflammation, pyrexia
	Not known	: Injection site reaction, malaise, face oedema, chills, oedema peripheral, chest pain
Investigations	Not known	: Hepatic enzyme increased

OVERDOSE AND TREATMENT

The clinical symptoms of metronidazole overdose are usually limited to nausea, vomiting and neurotoxic effects, including ataxia, slight disorientation, confusion, seizures and peripheral neuropathy. There is no specific treatment of overdose, metronidazole infusion should be discontinued. Patients should be treated symptomatically.

INCOMPATIBILITIES

Do not use equipment containing aluminum (e.g., needles, cannulae) that would come in contact with the drug solution as precipitates may form.

Metronidazole is incompatible with, but not limited to: aztreonam, cefamandole nafate, cefoxitin, and penicillin G. In the absence of compatibility studies, this medicinal product must not be mixed with other drug.

STORAGE CONDITIONS

Store below 30°C.

Protect from light.

SHELF LIFE

3 years.

DOSAGE FORMS AND PACKAGING AVAILABLE

Box containing: 1 vial of 100 mL sterile solution

DATE OF REVISION OF PI

Date of revision: 09/08/2024

Manufacturer :-
SIAM BHEASACH CO., LTD.
123 Soi Chokechainummitr, Vibhavadi-Rangsit Road, Chomphon, Chatuchak, Bangkok 10900 and 9 Soi Chokechainummitr 3, Vibhavadi-Rangsit Road, Dindang, Dindang, Bangkok 10400, Thailand.

Product Registration Holder, Importer :-
SPG PHARMA (MALAYSIA) SDN. BHD.
Suite 1701, Block A, Damansara Intan e-Business Park,
No. 1, Jalan SS20/27, 47400 Petaling Jaya, Selangor Darul Ehsan, Malaysia.

METR-INJ-500-03-L-ENG-MY-A0005

Copyright © 2024 Siam Bheasach. All rights reserved.

Actual Size 100 %